

V4.2 K_V42 Comparative Report

Purpose:
Calibrate the diagnostic ruler before a Dynamic Orbit Test.

$$K_{V42} = F_{in} + F_{bundle} - 2P_{core} - 2P_{boundary} - P_{shortcut}$$

F_{in} = inward movement of off-axis paths near the central source.
 F_{bundle} = narrowing/convergence of the whole path bundle behind the source.
 P_{core} = fraction of paths entering $r \leq 2.0$; penalizes wormhole/core collapse.
 $P_{boundary}$ = fraction of paths touching $|y| \geq 9$; penalizes wall/boundary fleeing.
 $P_{shortcut}$ = excessive cost reduction beyond allowed gain; penalizes magic tunnels.

Guardrail: K_{V42} is not Ricci curvature, not GR curvature, and not gravitational lensing.

Top cases by K_{V42} :

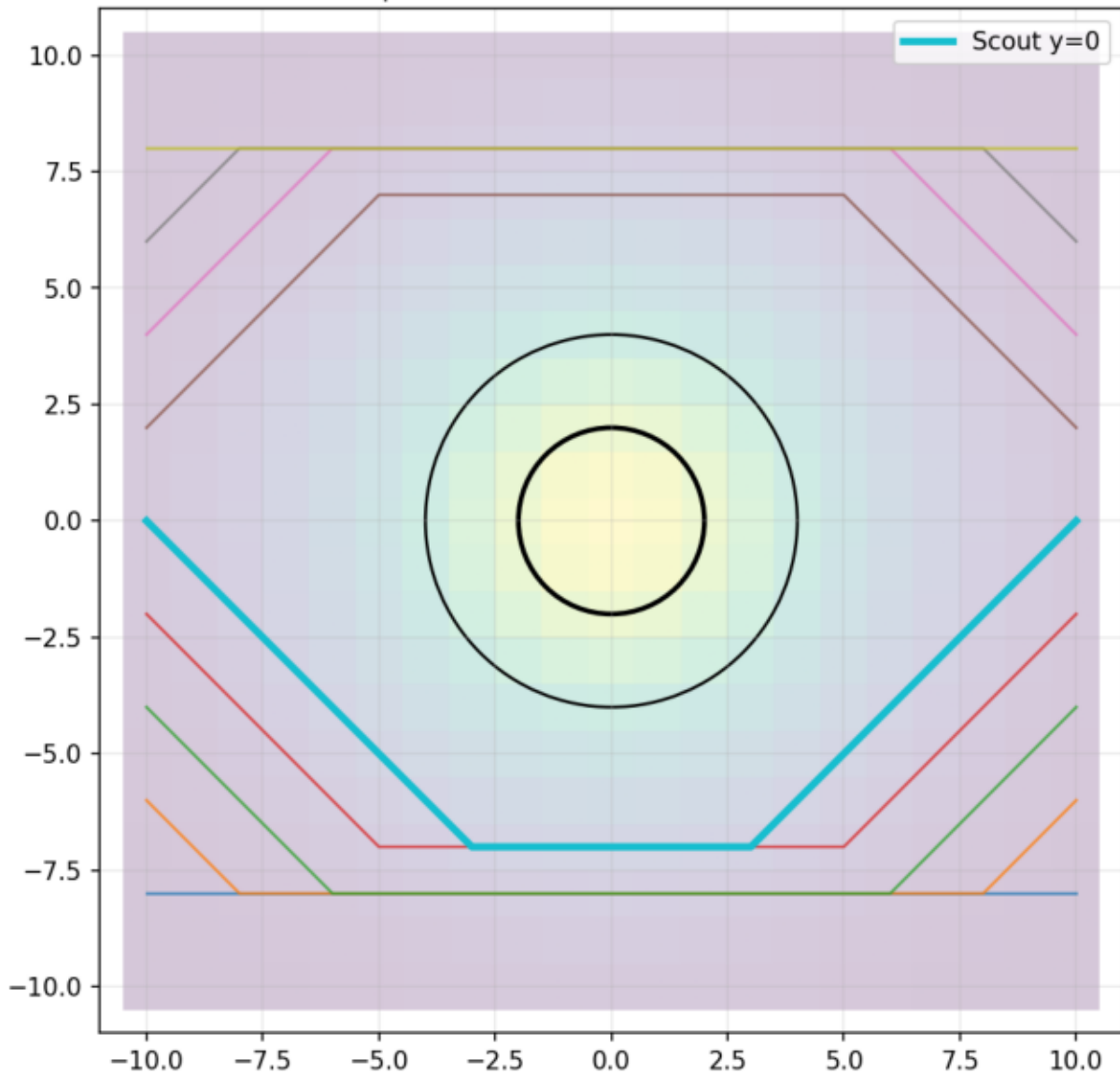
1D-C3_connect_mu2: $K=13.30$, $F_{in}=5.00$, $F_b=16.00$, $P_c=1.00$, $P_s=5.70$
1D-B_connect_mu5: $K=10.87$, $F_{in}=4.00$, $F_b=14.00$, $P_c=1.00$, $P_s=5.13$
1D-C3_lamr3_mu3: $K=7.87$, $F_{in}=2.50$, $F_b=8.00$, $P_c=0.56$, $P_s=1.52$
1D-C3_lamr2_mu3: $K=6.54$, $F_{in}=3.00$, $F_b=8.00$, $P_c=0.78$, $P_s=2.90$
1D-C3_lamr3_mu5: $K=6.34$, $F_{in}=3.75$, $F_b=10.00$, $P_c=1.00$, $P_s=5.41$
1D-C3_lamr4_mu6: $K=5.41$, $F_{in}=3.25$, $F_b=10.00$, $P_c=1.00$, $P_s=5.84$
1D-C1_l1_m3: $K=0.14$, $F_{in}=1.25$, $F_b=0.00$, $P_c=0.56$, $P_s=0.00$
1D-C3_lamr2_mu2: $K=-0.42$, $F_{in}=0.25$, $F_b=0.00$, $P_c=0.33$, $P_s=0.00$
1D-A_lambda0: $K=-0.67$, $F_{in}=0.00$, $F_b=0.00$, $P_c=0.33$, $P_s=0.00$
1D-B_flat: $K=-0.67$, $F_{in}=0.00$, $F_b=0.00$, $P_c=0.33$, $P_s=0.00$

First read: this v0 ruler must be judged against the plots. If it rewards a visually pathological case, the ruler itself must be revised.

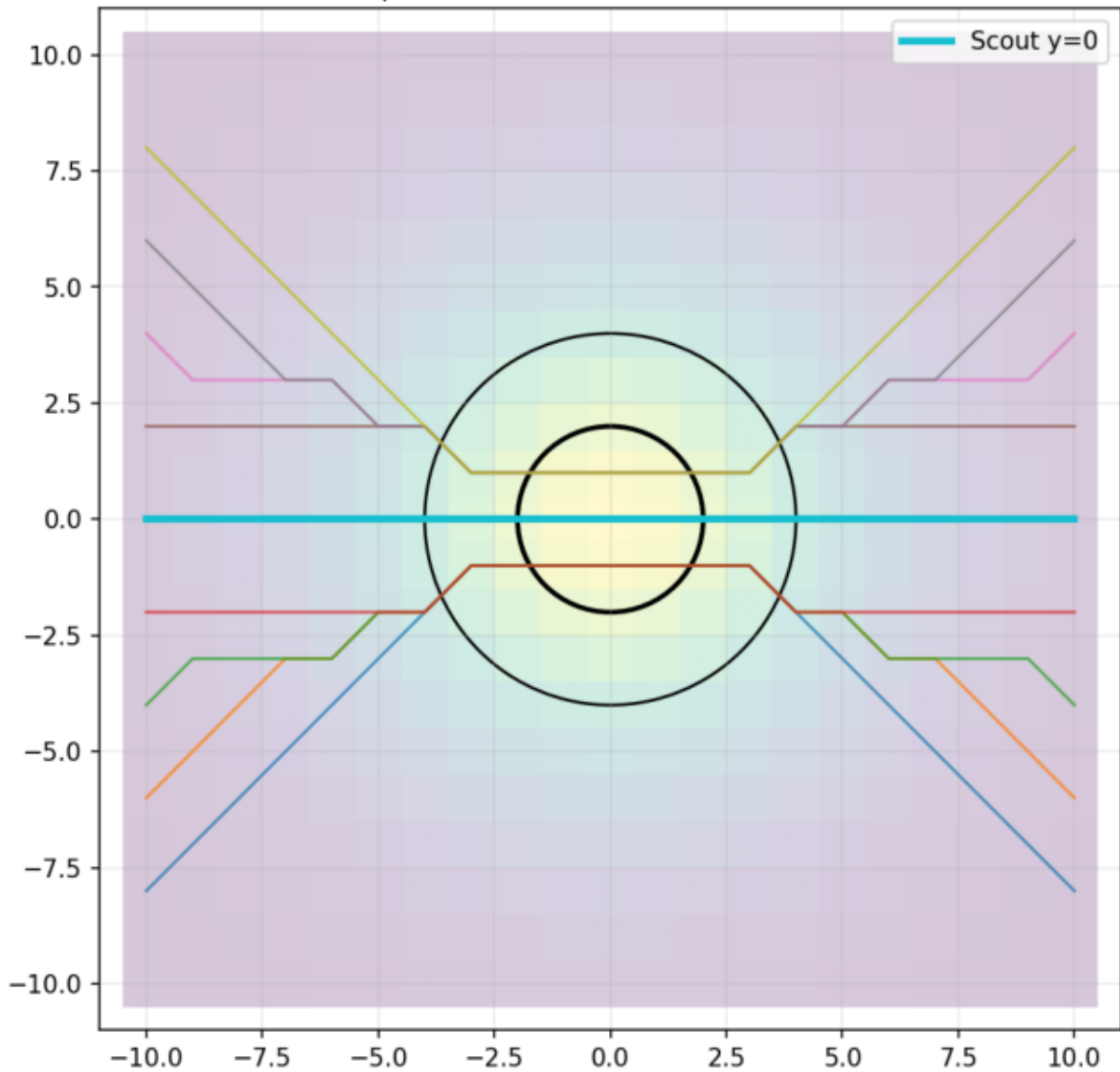
K_V42 Ranking Table - Top 20

01.	1D-C3_connect_mu2	K= 13.30	Fin= 5.00	Fb=16.00	Pc=1.00	Pb=0.00	Ps= 5.70
02.	1D-B_connect_mu5	K= 10.87	Fin= 4.00	Fb=14.00	Pc=1.00	Pb=0.00	Ps= 5.13
03.	1D-C3_lamr3_mu3	K= 7.87	Fin= 2.50	Fb= 8.00	Pc=0.56	Pb=0.00	Ps= 1.52
04.	1D-C3_lamr2_mu3	K= 6.54	Fin= 3.00	Fb= 8.00	Pc=0.78	Pb=0.00	Ps= 2.90
05.	1D-C3_lamr3_mu5	K= 6.34	Fin= 3.75	Fb=10.00	Pc=1.00	Pb=0.00	Ps= 5.41
06.	1D-C3_lamr4_mu6	K= 5.41	Fin= 3.25	Fb=10.00	Pc=1.00	Pb=0.00	Ps= 5.84
07.	1D-C1_l1_m3	K= 0.14	Fin= 1.25	Fb= 0.00	Pc=0.56	Pb=0.00	Ps= 0.00
08.	1D-C3_lamr2_mu2	K= -0.42	Fin= 0.25	Fb= 0.00	Pc=0.33	Pb=0.00	Ps= 0.00
09.	1D-A_lambda0	K= -0.67	Fin= 0.00	Fb= 0.00	Pc=0.33	Pb=0.00	Ps= 0.00
10.	1D-B_flat	K= -0.67	Fin= 0.00	Fb= 0.00	Pc=0.33	Pb=0.00	Ps= 0.00
11.	1D-B_equal_l1_m1	K= -0.67	Fin= 0.00	Fb= 0.00	Pc=0.33	Pb=0.00	Ps= 0.00
12.	1D-B_focus_l1_m2	K= -0.67	Fin= 0.00	Fb= 0.00	Pc=0.33	Pb=0.00	Ps= 0.00
13.	1D-B_delay_l2_m1	K= -0.67	Fin= 0.00	Fb= 0.00	Pc=0.33	Pb=0.00	Ps= 0.00
14.	1D-C1_flat	K= -0.67	Fin= 0.00	Fb= 0.00	Pc=0.33	Pb=0.00	Ps= 0.00
15.	1D-C2_flat	K= -0.67	Fin= 0.00	Fb= 0.00	Pc=0.33	Pb=0.00	Ps= 0.00
16.	1D-C2_lambda0_only	K= -0.67	Fin= 0.00	Fb= 0.00	Pc=0.33	Pb=0.00	Ps= 0.00
17.	1D-C3_flat	K= -0.67	Fin= 0.00	Fb= 0.00	Pc=0.33	Pb=0.00	Ps= 0.00
18.	1D-C3_lambda0_only	K= -0.67	Fin= 0.00	Fb= 0.00	Pc=0.33	Pb=0.00	Ps= 0.00
19.	1D-C3_core_delay_only	K= -0.75	Fin=-0.75	Fb= 0.00	Pc=0.00	Pb=0.00	Ps= 0.00
20.	1D-C1_connect_m3	K= -0.92	Fin= 3.00	Fb= 0.00	Pc=0.78	Pb=0.00	Ps= 2.37

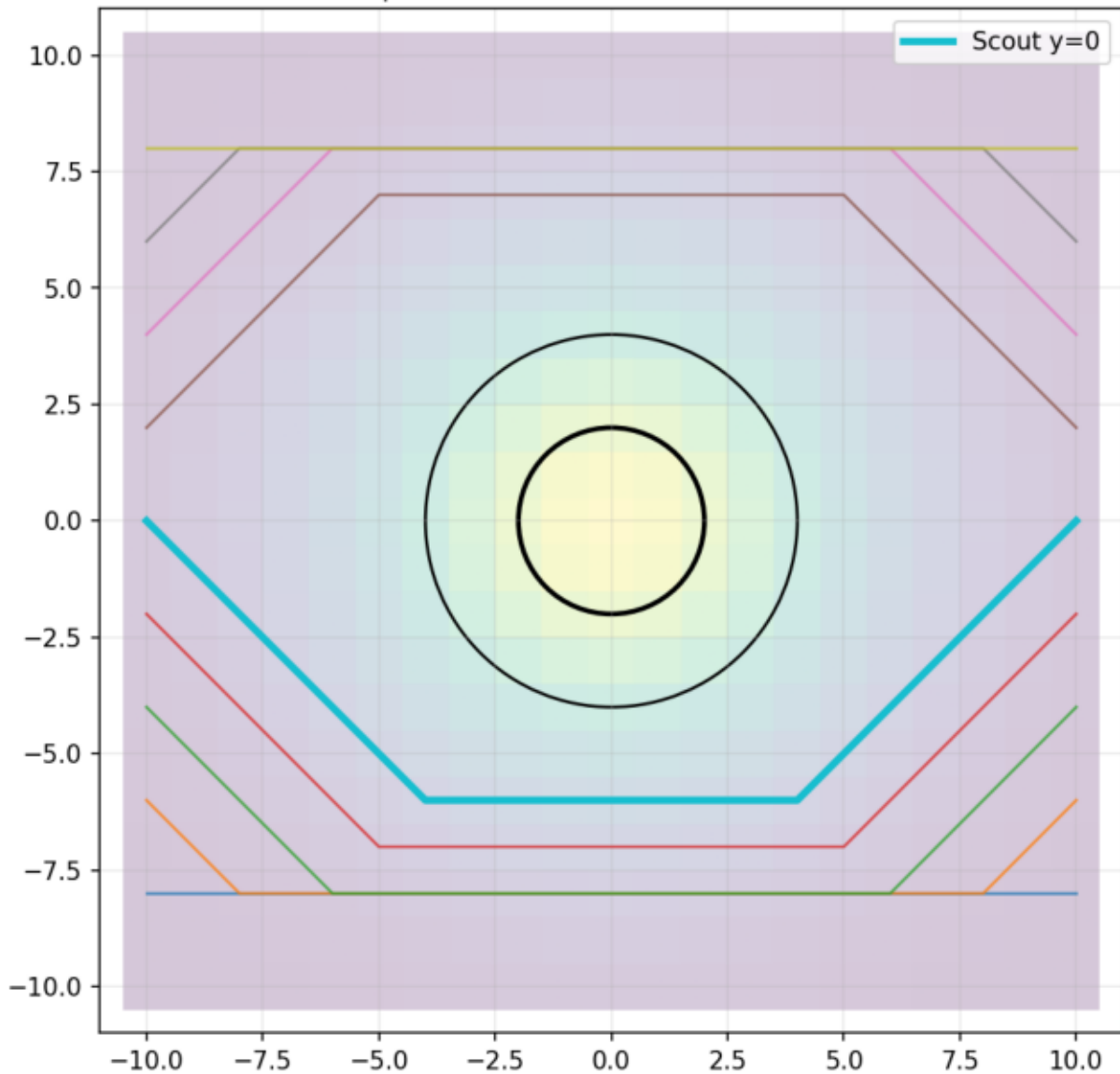
1D-A_{lambda10}
K=-2.75 | Fin=-2.75, F_b=0.00, P_c=0.00, P_s=0.00



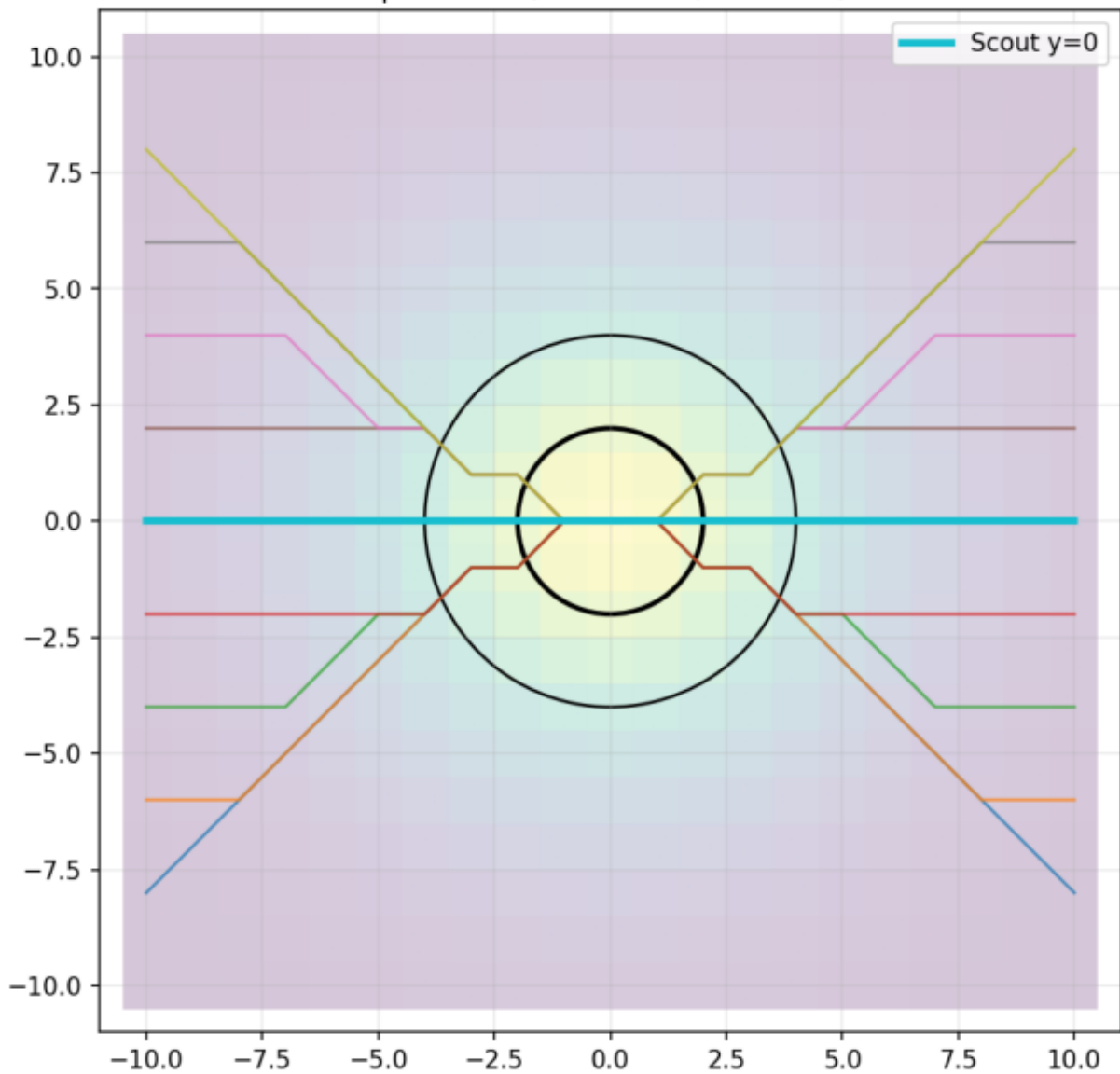
1D-B_connect_mu5
K=10.87 | Fin=4.00, Fb=14.00, Pc=1.00, Ps=5.13



1D-C2_delay_lamr3_mu1
K=-2.75 | Fin=-2.75, Fb=0.00, Pc=0.00, Ps=0.00



1D-C3_connect_mu2
K=13.30 | $F_{in}=5.00$, $\bar{F}_b=16.00$, $P_c=1.00$, $P_s=5.70$



1D-C3_lamr2_mu3
K=6.54 | Fin=3.00, Fb=8.00, Pc=0.78, Ps=2.90

